

Optimized Auto-Rickshaw Ride-Sharing for Campus Commuters

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Abstract—In urban environments, optimizing shared transportation is crucial for reducing costs and improving efficiency, especially for students. The approach involves transforming the city's map into a weighted tree structure based on frequently visited locations and estimated fares. Passengers' destinations and preferred timings are analyzed to compute trip similarity scores, facilitating the clustering of individuals with compatible routes and schedules. Dynamic programming techniques are employed to determine optimal seating arrangements when group sizes exceed vehicle capacity. Additionally, the model proposes optimal departure times by integrating individual time convenience functions of passengers in a rickshaw. This methodology aims to provide a cost-effective and efficient ride-sharing solution tailored to the specific commuting patterns of the IISER Bhopal Student Community.

Index Terms—ride-sharing optimization, combinatorial optimization, dynamic programming, clustering algorithms

I. INTRODUCTION

In today's dynamic ride-sharing environment, ensuring that each passenger pays a fair share of the total fare is a critical challenge. When groups with high route overlaps are identified, it becomes imperative to assign these groups to vehicles in an optimal manner. The central problem is to assign passenger groups to vehicles such that the maximum cost incurred per individual is minimized. Each group consists of a set of people who prefer to travel together, and when a group's size exceeds the vehicle's capacity, the group must be optimally split into smaller subgroups (or trips) to ensure that no one bears an excessively high cost. The ultimate objective is to distribute the cost of each segment proportionally among all passengers, so that every individual pays only for the segments they share, thereby minimizing the worst-case (maximum) fare per head.

Problem Statement

The problem involves:

- Assigning Passenger Groups to Vehicles: Groups must be allocated to vehicles with a fixed capacity.
- Optimal Group Splitting: If a group exceeds the vehicle's capacity, it must be divided into smaller trips.
- Fair Cost Allocation: The fare of each route segment is split equally among all passengers riding that segment.
- Minimizing the Maximum Individual Fare: The goal is to ensure that the highest fare any individual pays, over all shared segments, is as low as possible.

II. LITERATURE REVIEW

Ride-sharing optimization is a critical area of research aimed at improving efficiency, reducing costs, and enhancing user satisfaction in shared mobility services. Various approaches have been proposed in the literature to optimize route planning, passenger matching, and cost allocation. This section explores key studies and methodologies relevant to our project.

The paper "A Ride Sharing System for University Community"[8] by Olaniyan et al. proposes a campus-specific ride-sharing platform to reduce commuting costs, traffic congestion, and parking issues. It features ride-matching, trip advertisements, geo-location tracking, and user verification, ensuring efficient and secure ride coordination. The system encourages carpooling among students and staff, promoting sustainability while optimizing travel routes. With a user-friendly interface for booking and communication, this study provides a practical framework for implementing an affordable and efficient ride-sharing solution tailored to university environments.

The paper "Optimization for dynamic ride-sharing: A review"[4] by Niels Agatz, Alan Erera, Martin Savelsbergh, and Xing Wang provides a comprehensive survey of optimization techniques for dynamic ride-sharing systems. It discusses key challenges such as real-time passenger matching, route planning, and cost allocation while considering constraints like time windows, detours, and ride comfort. The authors explore various optimization approaches, including integer programming, heuristic algorithms, and simulation-based models, to improve efficiency. The study highlights the role of demand prediction, pricing strategies, and technological advancements in enhancing ride-sharing operations and reducing congestion. The paper concludes by emphasizing the need for scalable and adaptive algorithms to handle real-world complexities in dynamic ride-sharing.

The paper "Optimizing ride-sharing operations in smart sustainable cities: Challenges and the need for agile algorithms"[6] by Martins et al. explores the complexities of ride-sharing optimization in dynamic urban environments, emphasizing the need for agile algorithms to handle real-time demand fluctuations, traffic uncertainties, and multi-objective constraints like cost, travel time, and sustainability. It advocates for hybrid metaheuristic and AI-driven approaches,

including reinforcement learning and deep learning, to enhance passenger matching and route optimization. The study also highlights the role of electric and autonomous vehicles, multi-modal integration with public transport, and real-time pricing models in making ride-sharing more efficient and eco-friendly.

The paper "Distance Browsing in Spatial Databases"[2] by Hjaltason and Samet introduces an incremental nearest neighbor (INN) search algorithm, which efficiently retrieves spatial objects one by one using R-trees, reducing redundant computations compared to traditional k-NN search. This method enables dynamic distance browsing and is particularly beneficial for ride-sharing platforms, where it can optimize real-time ride matching, surge pricing, carpooling, ETA calculations, and location-based promotions. By implementing INN, ride-sharing apps can achieve faster, more efficient driver-rider matching, enhancing user experience while minimizing computational overhead.

The paper "User Preferences in Ride-Sharing Mathematical Models for Enhanced Matching" [9] introduces a mathematical programming model that incorporates user preferences into ride-sharing scenarios, aiming to improve both efficiency and user satisfaction. A key feature is the consideration of passenger transfers, allowing individuals to switch vehicles at designated meeting points, which is particularly beneficial in less-populated areas. The authors propose a Preference-Driven Matching Algorithm to effectively capture and integrate user preferences during the matching process. Additionally, they develop an Iterative Enhance-and-Optimize Algorithm designed to produce high-quality solutions with reduced computational times. Evaluations across various real-scale scenarios demonstrate that incorporating user preferences leads to higher satisfaction rates, more responsive systems with shorter response times, and increased demand and revenue by servicing more users. The study underscores the importance of balancing efficiency with user satisfaction in optimizing ride-sharing systems.

The paper "A Genetic Algorithm Approach to Multi-Objective Scheduling Problems with Regular and Non-Regular Objective Functions"[1] explores the use of Genetic Algorithms (GA) for multi-objective scheduling in identical parallel machine scheduling, optimizing both makespan and earliness/tardiness penalties. GA is employed to generate Pareto-optimal schedules by solving job assignment, sequencing, and timing subproblems. A key application of this approach is in ride-sharing platforms, where it can optimize driver allocation, balancing customer wait time and driver earnings, and route scheduling, minimizing detours while ensuring timely arrivals. GA can also improve carpooling efficiency by generating optimal shared ride schedules, optimize dynamic pricing to balance revenue and demand, and enhance fleet management by ensuring fair workload distribution among drivers. By implementing GA-based scheduling, ride-sharing platforms can achieve greater efficiency, fairness, and an improved user experience.

The paper "Real-time city-scale taxi ride-sharing"[5] proposes an efficient algorithm for dynamically matching pas-

sengers with taxis while minimizing detours and maximizing shared rides. Using real-time GPS data, historical demand, and traffic conditions, the system optimizes ride assignments through a search-based matching algorithm that ensures minimal computation time. The study highlights scalability, profit-driven matching, and route optimization, making it highly relevant for designing a campus ride-sharing system that efficiently schedules rides based on student travel patterns.

The paper "Cost-sharing mechanism design for ride-sharing"[7] presents a mathematically rigorous and incentive-compatible cost-sharing framework for dynamic ride-sharing. It focuses on fairness, stability, and efficiency, ensuring that each passenger pays a reasonable amount based on their contribution to the shared ride. The authors propose game-theoretic models to prevent strategic manipulation and ensure passengers have no incentive to misreport their trip details.

Study	Car-Sharing Problem		Methodology		
	Static	Dynamic	Exact	Heuristic or Metaheuristic	Other
Kornhauser et al. (1977)		•			•
Baldacci et al. (2004)	•		•	•	
Agatz et al. (2011)		•	•		
Yan and Chen (2011)	•			•	•
Herbawi and Weber (2012)		•		•	
Hosni et al. (2014)	•	•	•	•	
He et al. (2014)	•				•
Lee and Savelsbergh (2015)		•	•	•	
Fagnant and Kockelman (2015)		•		•	•
Santos and Xavier (2015)	•	•		•	
Huang et al. (2015)	•		•	•	
Naoum-Sawaya et al. (2015)	•		•		
Stiglic et al. (2015)	•		•		
Schreieck et al. (2016)		•			•
Jung et al. (2016)		•		•	
Alonso-Mora et al. (2017)		•	•	•	
Levin et al. (2017)	•	•		•	
Masoud and Jayakrishnan (2017)	•		•		
Najmi et al. (2017)	•	•		•	
Wang et al. (2017)		•	•	•	
Li et al. (2018b)	•		•	•	
Long et al. (2018)	•				•
Ma et al. (2018)	•			•	
Lokhandwala and Cai (2018)	•			•	•
Yu et al. (2019)	•		•		
Chen et al. (2019)	•		•	•	
Simonetto et al. (2019)		•		•	
Li and Chung (2020)	•		•	•	
Cheikh-Graïet et al. (2020)		•		•	

Figure 1: Literature review

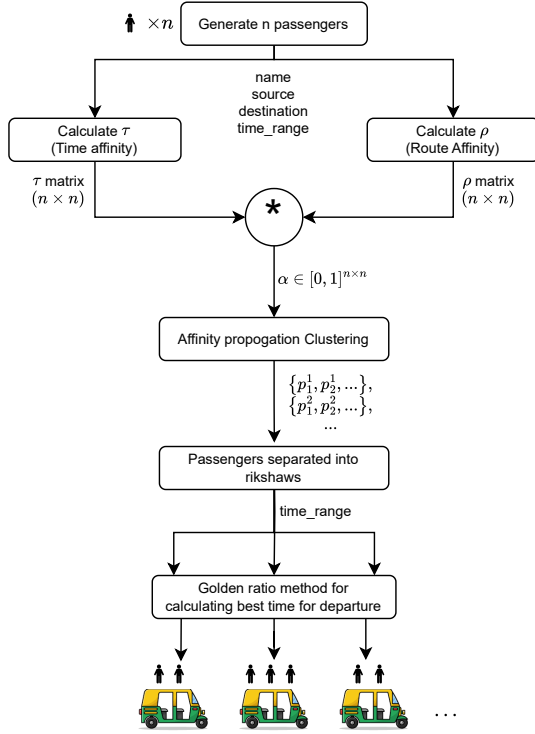


Figure 2: Flowchart for Yatry

Figure 1 categorizes some more car-sharing studies by static vs. dynamic approaches and methodologies used. It highlights the shift towards dynamic ride-sharing and the reliance on heuristic/metaheuristic methods for solving complex optimization problems. These insights help in selecting suitable approaches for our campus ride-sharing project.

III. METHODOLOGY

A. Overview

The proposed methodology involves a stage-wise approach to solving the ride-sharing combinatorial optimization. This includes two main stages and one conditional stage.

- 1) Route-based affinity scoring
- 2) Timing-based affinity scoring
- 3) Multi-vehicle passenger distribution (*conditional*)

Such a stage-based approach was chosen as it was intuitive to first group passengers based on their journey routes and then on their preferred times of departure as it does not make sense to search for passengers with similar time preferences among the set of those who have dissimilar trips. In first two stages, corresponding “affinity” or “compatibility” scores are obtained between every pair of passengers. Based on these affinities, passengers are clustered into groups that can share a vehicle. If the group size exceeds the maximum capacity of a vehicle, a multi-vehicle passenger distribution algorithm optimizes savings for the group by sharing the vehicle.

Several assumptions and simplifications were made to tailor the algorithm to the scale and pattern of commute of students

in medium sized cities, *e.g.*, students of IISER Bhopal. Further, the application of heuristic and stochastic methods to get to ride-sharing solutions which are reasonably optimal.

B. Route-Affinity Scoring

To find the routes for each passenger, the map of the area needs to be modeled as a graph, with nodes representing individual locations and edges presenting the road between them. In our simplified model of the city, only the locations which are very frequently traveled to by the students are considered. Also, to simplify the problem of selecting one of multiple routes for each passenger, the graph is modeled as a tree, thus satisfying the property that there is only one path between every pair of nodes in the graph. Since the passengers’ routes would always have the institute (here, IISER Bhopal) as either the source or the destination, the institute is fixed as the root of this tree-map.

Let passenger P_n follow the route on the map consisting of the sequence of k_n nodes in order $r_n = \langle a_1^{(n)}, a_2^{(n)}, \dots, a_{k_n}^{(n)} \rangle$, where the said passes through k nodes. The route affinity score of P_i for P_j is given by

$$\rho_{ij} = \begin{cases} \frac{f_{\text{fare}}(\text{LCP}(r_i, r_j))}{f_{\text{fare}}(r_i)}, & \text{if } \text{LCP}(r_i, r_j) \in \{r_i, r_j\} \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where $\text{LCP}(r_i, r_j)$ is the longest common prefix between the two routes, r_i and r_j and $f_{\text{fare}}(r)$ denotes the fare charged to travel along route r . The asymmetry in this function arises from the fact that a longer common route results in a higher amount of fare-sharing, and thus more savings. The condition ensures that the routes do not diverge at any point.

C. Time Convenience Distribution Modeling

To model a passenger’s preferred time of departure, we assume that their time preference lies between a minimum ($t_{\min}$) and maximum ($t_{\max}$) preferred time. We model this preference using a normal distribution such that a proportion m_{range} (typically 0.8) of the total probability mass lies within this interval.

Mean and Standard Deviation Calculation

The mean μ of the distribution is the midpoint of the time window:

$$\mu = \frac{t_{\min} + t_{\max}}{2}$$

To determine the standard deviation σ , we compute the z -score corresponding to the desired central probability mass:

$$z = \Phi^{-1} \left(\frac{1 + m_{\text{range}}}{2} \right)$$

$$\sigma = \frac{t_{\max} - t_{\min}}{2z}$$

Here, Φ^{-1} denotes the inverse cumulative distribution function (percent-point function) of the standard normal distribution.

D. Time Affinity Score Between Passengers

To evaluate the compatibility of time preferences between two passengers, we calculate a time affinity score. This score is based on how much of the first passenger's time preference distribution lies within the second passenger's preferred time window.

Let passenger 1's preference be modeled as a normal distribution with mean μ_1 and standard deviation σ_1 , and let passenger 2 prefer departures between t_{2_min} and t_{2_max} .

Affinity Score Calculation

We compute the affinity score as the probability mass of passenger i 's distribution that falls within passenger j 's preferred window:

$$\tau_{ij} = \min \left(1, \frac{\Phi_i \left(\frac{t_{max}^j - \mu_i}{\sigma_i} \right) - \Phi_i \left(\frac{t_{min}^j - \mu_i}{\sigma_i} \right)}{m_{range}} \right) \quad (2)$$

where Φ_i is the cumulative distribution function (CDF) of the time convenience function for passenger i .

This score ranges from 0 (no overlap) to 1 (perfect alignment) and provides a probabilistic measure of time compatibility between the two passengers.

E. Affinity-based Clustering

Three affinity matrices are constructed to denote the affinity of passenger i to share a ride with passenger j based on different conditions:

- 1) The time affinity matrix $\tau \in [0, 1]^{n \times n}$
- 2) The route affinity matrix $\rho \in [0, 1]^{n \times n}$
- 3) The combined affinity matrix $\alpha \in [0, 1]^{n \times n}$

τ and ρ are calculated from Equation 2 and Equation 1 as

$$\rho = \begin{bmatrix} \rho_{11} & \rho_{12} & \dots & \rho_{1n} \\ \rho_{21} & \rho_{22} & \dots & \rho_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{n1} & \rho_{n2} & \dots & \rho_{nn} \end{bmatrix} \quad (3a)$$

$$\tau = \begin{bmatrix} \tau_{11} & \tau_{12} & \dots & \tau_{1n} \\ \tau_{21} & \tau_{22} & \dots & \tau_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \tau_{n1} & \tau_{n2} & \dots & \tau_{nn} \end{bmatrix} \quad (3b)$$

$$\alpha = \begin{bmatrix} \alpha_{11} & \alpha_{12} & \dots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \dots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \dots & \alpha_{nn} \end{bmatrix} \quad (3c)$$

where

$$\alpha_{ij} = \tau_{ij} \rho_{ij}$$

and min-max scaling is applied on α to scale values for ease in clustering. α is an asymmetric matrix.

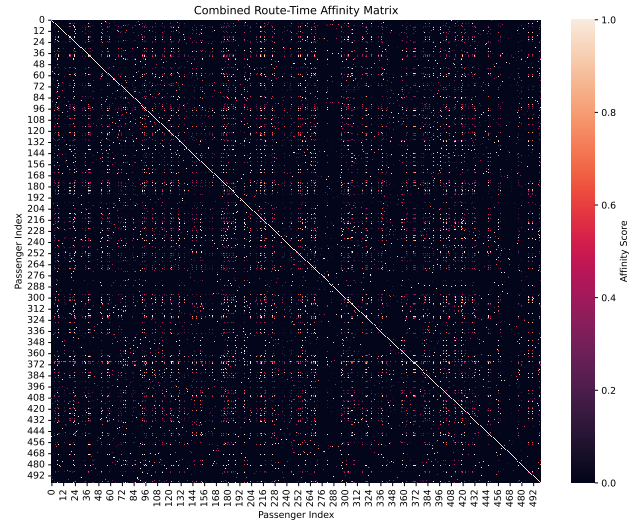


Figure 3: Combined affinity matrix α for 500 random passengers. Bright spots show passengers who are compatible for ride sharing.

For clustering, we convert the affinity matrix α to a matrix containing a distance metric between passengers i and j , given by

$$\delta_{ij} = 1 - \alpha_{ij}$$

and the distance matrix $\delta \in [0, 1]^{n \times n}$ gives the distances between different passengers. We use the affinity propagation clustering algorithm [3] to find clusters of passengers who can conveniently share a ride.

IV. RESULTS

Detail	Value
Optimized Departure Time	09:09 24 Apr 2025
Total Passengers	7
Total Fare	200.00
Total Savings	1000.00

V. CONCLUSION AND FUTURE WORK

Ride-sharing among communities for daily commute across a geographical region is a temporally and financially beneficial practice, which is also good for the environment and reduces pollution and carbon footprints. Optimizing this practice leads to higher monetary savings for the community members, especially valuable for student communities without personal transportation resources. Also, it makes it easy to find people to share a ride with and encourages people to travel, leading to more economic growth across the nearby region, as students and the youth are the most thriving and populous age group in the country.

On the technical side, we plan to collect anonymous data from users, preserving their privacy. This data can be used to build heuristics to predict patterns of travel based on

Passenger	Route	Original Fare	New Fare	Savings	Savings (%)
#126	IISER Bhopal → Bairagarh	200.00	33.33	166.67	83.33
#143	IISER Bhopal → Shivhare	100.00	16.67	83.33	83.33
#258	IISER Bhopal → Bairagarh	200.00	33.33	166.67	83.33
#410	IISER Bhopal → Bairagarh	200.00	33.33	166.67	83.33
#454	IISER Bhopal → Bairagarh	200.00	33.33	166.67	83.33
#467	IISER Bhopal → Bairagarh	200.00	33.33	166.67	83.33
#490	IISER Bhopal → Shivhare	100.00	16.67	83.33	83.33

Table I: Sample grouping of different passengers and their fares

dates, times and locations to help users share rides more easily. We could also formulate the problem as a Markov Decision Process to train Deep RL algorithms to optimize across multiple possible routes between points, lifting the single route constraint discussed before.

We plan to implement the ride sharing optimization algorithm as a mobile and web application, facilitating ease of use by the IISER Bhopal student community and later scale to other similar communities as well. Migrating students from manual methods of planning ride-sharing such as WhatsApp groups and other platforms lets people plan more trips without worrying about finding people to share a ride with, which is usually cumbersome and time taking, leading to frequent cancellation of plans. More trips, more memories, more recreation, and everyone is happy! In conclusion, ingenious and community-driven algorithms like ride-sharing are highly valuable to society and the environment, and optimizations provide exponential benefits and brings more inclusivity, networking and overall prosperity to the community.

VI. IMPLEMENTATION SUMMARY

The implementation phase of this project involved the development of a comprehensive system for optimizing shared auto-rickshaw rides in Bhopal, with a specific focus on leveraging both spatial and temporal passenger preferences. The system architecture was built using Python and modularized into separate components for map creation, data preprocessing, affinity scoring, and clustering.

To begin with, a transportation map of Bhopal was modeled as a tree graph using custom `Map` and `Location` classes. Each node in the tree represents a real-world location (such as IISERB, Lal Ghati, Bhopal Junction, etc.), and each edge is annotated with a corresponding rickshaw fare, effectively encoding the cost of travel between locations. The tree was rooted at IISERB, which serves as the central hub for all operations. All locations were registered as part of the map, and directed edges were added between significant points in the city to reflect practical rickshaw routes.

Upon receiving source and destination inputs from passengers, a path-searching algorithm was employed to determine the optimal travel path. This algorithm leverages the tree structure to find valid routes from or to IISERB, ensuring that the trip is feasible and cost-effective. Only routes that include IISERB as either the source or the destination are considered, aligning with the real-world constraint that the service is anchored at the IISER campus.

Passenger data, including their selected routes and preferred travel time windows, were then processed to derive two critical affinity measures: route affinity and time affinity. Route affinity between two passengers was computed based on the degree of overlap in their travel paths. This was quantified by comparing the sets of intermediate nodes visited during each passenger’s journey. A greater overlap implies a higher likelihood of shared travel feasibility.

For time affinity, each passenger’s temporal preference was modeled as a Gaussian distribution, centered around their preferred departure time and spanning the interval defined by their earliest and latest acceptable departure times. The similarity between these distributions for any pair of passengers was measured using the Hellinger distance—a symmetric, bounded, and probabilistically interpretable metric. This ensured that the temporal compatibility score was normalized between 0 and 1 and suitable for integration with other affinity scores.

After calculating both the route and time affinity matrices, these were element-wise multiplied to form a composite affinity matrix capturing the joint compatibility of passengers. This matrix was then scaled using Min-Max normalization to ensure all values fell within the same numerical range, preventing any metric from disproportionately influencing clustering outcomes.

Finally, Affinity Propagation clustering was applied to the normalized matrix. This unsupervised algorithm grouped passengers into clusters based on the strength of mutual affinities, resulting in optimal ride-sharing groups. Post-clustering, the system calculated the fare per passenger by dividing the total route fare across group members. Savings were then computed for each individual by comparing the cost of shared travel with the fare of a solo trip. This comprehensive implementation not only minimized travel costs but also respected the individual constraints of time and route, showcasing the viability of an intelligent auto-sharing system tailored to urban mobility scenarios.

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